Brayden Tearle

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Computer Engineering student with a Nuclear Engineering minor and hands-on experience in embedded systems, research instrumentation, and hardware/software integration. Interested in instrumentation and control systems for nuclear energy applications.

EDUCATION

Kennesaw State University – Marietta Campus
Bachelor of Science in Computer Engineering (GPA: 3.52) May 2028 (Expected)
Minor in Nuclear Engineering

RELEVANT COURSEWORK

Advanced Embedded Design	Spring 2027	**Computer Organization & Interfacing**	Spring 2027
Fundamentals of Nuclear Engineering	Spring 2027	**Engineering Electronics**	Spring 2027
Hardware Programming	Summer 2026	**Digital Logic Design with Lab**	Fall 2025
Circuit Analysis I with Lab	Spring 2026	**Physics II with Lab**	Fall 2026
Programming and Problem-Solving	Fall 2026	**Discrete Mathematics**	Fall 2026
Probability & Data Analysis	Spring 2026	**Ordinary Differential Equations**	Spring 2026

RESEARCH EXPERIENCE

Undergraduate Researcher / NESEL, Kennesaw State University **Fall 2026 – Present**

- Reverse-engineered an undocumented radiation detection and laser-marking system (Raspberry Pi, ESP32, stepper motors), reconstructing its architecture from minimal documentation
- Mapped how the Python/Flask web interface, ESP32 serial pan/tilt link, and GPIO stepper control integrate
- Redesigned and 3D printed the laser module housing in Autodesk Fusion, correcting flaws in the original design that affected laser mounting and alignment
- Developing web-interface software for the system, starting with a live crosshair overlay showing where the laser is aimed

PROJECTS

Space Adventure Game – STM32 Cortex-M4, C, ARM Assembly

- Developed a multi-chapter embedded text adventure in C running on an STM32 Cortex-M4 microcontroller, using UART serial as the game console for player input and output
- Built puzzles that compute answers with inline ARM assembly (MOV, ADD, LSL) and applied concepts such as bit widths and Ohm's law
- Set up dual-target builds via conditional compilation: x86 for fast testing/debugging, ARM via arm-none-eabi-gcc and a linker script

Traffic Light Sequencer – 555 Timer, 74LS163 Counter, Digital Logic

- Built a traffic-light controller clocked by a 555 timer, using a 74LS163 4-bit synchronous counter and decoding logic to drive the light sequence

TECHNICAL SKILLS

Programming: Python, C, ARM Assembly, MATLAB
AI / LLM Tools: Claude, ChatGPT, Copilot (AI-assisted coding, debugging, and documentation)
Hardware: STM32, Raspberry Pi, ESP32, stepper motors, UART/serial, GPIO, Boolean logic and sequential circuits
Tools: Multimeter, Oscilloscope, LTspice, Autodesk Fusion, STM32CubeMX, Visual Studio Code, Linux, Flask, Microsoft Office

PROFESSIONAL EXPERIENCE

Starbucks Kennesaw, GA
Shift Supervisor, Barista Trainer, and Barista November 2020 – Present

- Lead shifts and train baristas to maintain efficiency, product quality, and customer satisfaction
- Developed standard operating procedures to improve team efficiency and effectiveness

CAMPUS & COMMUNITY INVOLVEMENT

Member / KSU Nuclear Society **January 2026 – Present**

- Engage with nuclear industry professionals and peers through technical seminars, networking events, and outreach

Member / Owls Formula Network **Fall 2024 – Present**

- Engage with fellow students in discussions and events on Formula 1 engineering, aerodynamics, and race technology

HONORS AND AWARDS

HOPE Scholarship Recipient August 2024 – Present
President's List Spring 2026